

FASTER POWER SYSTEM DESIGNING WITH ONLINE TOOLS

With the popularisation of computer-aided design (CAD) in mechanical designs, it soon encroached into other domains of electronics as well. The Internet further promoted it and now we have online e-CAD tools making news. Now electrical designers have tools to automate power electronics. These tools provide system designers a flexibility to architect and optimise end-to-end power systems.

Complex and compact electronic products and systems have driven demand for smaller, efficient and cost-effective power systems. Power designers have been forced to analyse bewildering topologies, architectures and myriads of products. Each alternative design approach requires performance calculations, data sheet searches, manual simulations, thermal projections, layout and packaging studies, bill-of-material (BOM) generation and cost analysis. To address this challenge, a variety of power design tools have been developed.

Updating to online system designs

The ability to work on the cloud has been the motivation behind many software systems taking the online road. And with this feature entering the power systems industry as well, we can expect widespread popularisation. Advances in internet coverage are expected to improve this situation further, but let's take a look at the web based power system designer first.

Exploring the Vicor's Power System Designer

The Vicor's Power System Designer (PSD) is a web-based tool for generating power systems fast—up to 75% faster than traditional methods. The Vicor's Power System Designer is the embodiment of the power component design methodology. The approach towards timely, low-resource, low-risk and high performance power system design called the power component design methodology (PCDM) provides the means to assemble optimal components to meet the demands of your power system design.

Libraries covers the required components, from power sources (AC and DC)

to the point of load, needed to build a high performance power system. The Vicor's PSD uses proven power components that are engineered to interface with each other thus reducing complexities of system design.

How to use the Vicor's PSD

Once you open the website, you are asked to enter your required design specifications and then the tool provides recommended designs based on some key figures of merit like the system efficiency, power footprint, component count, recommended best fit and cost. It provides a design environment that enables the designer to evaluate the complete power system designed, save it and then modify it.

Some hands-on experience

To get started, you need some power measurements for your system like the power source, voltage levels, number of outputs and the parameters for outputs. Once these specifications are entered into the tool, the Vicor's PSD delivers up to five, optimized designs based on the key figures of merit.

You can view the solutions in the HUB and analyse them based on your preferred figures of merit.

- System efficiency
- Power footprint
- Cost
- Component count
- Recommended best fit

You can drill down to the BOM to further verify the options and trade-offs. The solutions can also be viewed as a whiteboard.

HOW IT HAPPENS?

Power system modelling is not a simple task. Two distinct stages in designing are the design stage where an electric system model is created and the simulation stage where software simulation program are used to test the integrity of the design. Once the design has been successfully tested for real world operations by checking for specific types of design or operational problems, it is taken to the next step of implementation.

SUPPORTED COMPONENTS

Component families that support the power system designer

- ▶ PFM: Isolated AC-DC Converter with PFC
- ▶ AIM: AC Input Front-end
- ▶ ZVS Isolated DC-DC Converter
- ▶ DCM: DC-DC Converter Module
- ▶ BCM: Bus Converter Module
- ▶ IBC: Intermediate Bus Converter
- ▶ VTM: Current Multiplier
- ▶ PRM: Buck-Boost Regulator
- ▶ ZVS Buck Step-Down Regulator
- ▶ ZVS Buck Boost Regulator
- ▶ NBM: Non-Isolated Fixed Ratio Converter Module

The WhiteBoard provides a workspace to architect and analyse the power efficiency of your design requirements. You may explore each design to better understand the best fit for your design along with evaluating each element of the power chain. After analysing each power chain, you may select a solution and open a White Board for further evaluation and design. Final step is the creation of a BOM for easy ordering of the design.

Other features of the Vicor's PSD tool include setting the operating condition for each component of the power design to match the intended application and performance efficiency as well as loss analysis of individual component.

Why Vicor's PSD?

Vicor's PSD is not just a nice-to-have product selection tool. It is a robust, get-to-market faster time-saving tool! It sifts through data sheets, performing hundreds of calculations on the overall system performance, pulls together thousands of data points and draws a block Furthermore, it delivers a drawn-to-scale model of the power system along with the BOM. The Vicor's Power System Designer is a revolutionary new, web tool that simplifies and expedites the way power systems are designed.

To know more about Vicor Power, visit: <http://www.vicorpower.com>